Review Article

Internet of things in assistive technology for people with visual impairment: A scoping review

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Assistive technology (AT) can significantly enhance functional abilities among persons with disabilities. Recently, a rapid advance in computer and Internet of Things (IoT)-based AT has been observed with the integration of Artificial Intelligence. The present scoping review explores a range of IoT-based AT for vision impairments, focusing on their functioning, reliability, and identifying the existing gaps and opportunities for future enhancement. We used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. The search was conducted for the articles published between 2020 and March 2025. After the initial screening of both titles and abstracts of 705 articles, 18 full-text articles were selected for the review. It was found that the majority of the papers (38%) are related to smart canes/sticks. The findings indicate that several studies use Raspberry Pi and Arduino platform to make products cost-effective and flexible options for real-time processing. The accuracy of these devices ranges from 87.8% in navigation to 93% in voice control and 100% in object identification. Further, challenges such as usability testing, hardware limitations, dataset constraints, and environmental adaptability remain a matter of concern which need to be addressed in future innovation. Incorporating IoT technologies into vision-assistive devices has substantially improved accessibility. Future research should focus on enhancing real-time performance, AI-driven decision-making, and user-centric designs.

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The World Health Organization (2023) estimates that approximately 2.2 billion people worldwide are living with either near or distance visual impairment, while nearly half of these individuals have yet to be addressed.[1] In India, an estimated 4.95 million people are blind, and 70 million are visually impaired, including 0.24 million blind children.[2] Visual loss significantly affects the quality of life and inclusion and imposes a considerable economic and financial burden on individuals and their families. For example, the estimated annual net loss of gross national income due to blindness in India is around INR 845 billion. The per capita net loss of gross national income per blind person is INR 170,624.[2] These individuals with vision loss need one or more vision assistive technologies (ATs) to enhance performing daily living tasks and improve social inclusion and participation and the economy. AT also enables them to improve health care access for persons with disabilities, support an independent and productive life, obtain an education, and participate in the workforce and society by overcoming limitations and restrictions.[3] Such technology varies from basic and affordable to advanced, high-tech, and expensive devices.

In recent years, computer-based assistive technologies have significantly advanced, providing a wide range of support to people with different types of disabilities, including vision loss particularly in education and communication. For example, screen readers, such as Job Access with Speech (JAWS), Non-Visual Desktop Access (NVDA), and Apple's built-in Voice Over, are widely used by individuals with visual impairments to convert on-screen text into speech.[4] Among assistive technologies, the Internet of Things (IoT)-based ATs are becoming more prevalent as off-the-shelf technologies and are used by persons with disabilities.

Internet of Things

The IoT refers to a network of physical objects that are connected to the Internet, allowing them to interact and exchange data between users and devices. These 'things could range from an automobile equipped with built-in sensors to wearable healthcare monitor devices. The sensors embedded in these devices, along with the software applications they communicate with, enable them to monitor and respond to both internal states and external environmental conditions.[5] IoT minimizes the need for human intervention by enabling devices and objects to observe, identify, and understand situations autonomously.

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Although IoT technologies have shown strong potential to support people with visual impairment, several technical and accuracy related challenges still exist. Many IoT-based assistive devices depend on sensors such as ultrasonic, infrared, or camera modules to detect obstacles and guide users. However, these sensors often lose accuracy in difficult environments, for example, in poor light, rain, or crowded areas.[6] Delays in data transfer or weak processing power in low-cost IoT systems can also slow down responses, which may reduce the safety and reliability of real-time navigation.[7,8] Overall, there is a need for improvement in the accuracy, reliability, and real-world testing of IoT-based ATs.

Objective and research questions

The primary objective of this scoping review is to identify IoT-enabled vision ATs and to examine how these devices function and their reliability concerning assistive devices for vision [Table 1].

Methodology

The present scoping review was conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [Fig. 1]. The review examined: 1. the available studies related to IoT-enabled AT, 2. the mechanisms employed in functioning, 3. studies related to accuracy and reliability, and, finally, 4. gaps in IoT devices and future scope for improvement.

Search strategies

Relevant articles were gathered by searching PUBMED, Institute of Electrical and Electronics Engineers (IEEE) Xplore, Science Direct (SD), and Association for Computing Machinery (ACM) Digital Library using keywords such as "Assistive Technology," "Visually impaired," "IoT," and "assistive devices for the blind." Articles published from 2020 to 2025 were retrieved. We used logical operators like OR and AND and a search string through digital libraries. Interactive tests were carried out on each of the databases to identify the search strings whose returns were the most significant in terms of scope and relevance of the studies. Table 2 presents the database and keywords used and the respective search strings defined after the iterative validation process.

Table 1: Research questions (RQ)

ID	Question
RQ1	What vision-assistive products are available based on the IoT technology?
RQ2	How does the IoT-enabled vision-assistive technology function?
RQ3	What is the reliability of the IoT-enabled AT for vision?
RQ4	What should be the direction for future research for IoT-enabled AT for vision?

Study selection criteria

The purpose of this section is to provide a detailed description of the selection process, both inclusion [Table 3] and exclusion criteria [Table 4], used in this review paper. In applying the selection criteria, the articles were excluded if one of the exclusion criteria was met.

Data extraction

The data extraction process makes use of data extraction from generated sources, especially for this scoping review, using the Parsifal tool. Completing this field of this form after reading each selected article enables the extraction of data necessary to answer the research questions posed by this review paper.

Quality assessment

This section presents the quality assessment process used in this scoping review. The previous step produced a list of preselected articles. Each study was evaluated to ensure the quality of the data employing Parsifal tool developed by De Freitas *et al.*, 2022[9] [Table 5]. Each question has three options as answers, and each option has its respective score. Options are "yes," "partially," and "no," scoring 1.0, 0.5, and 0, respectively. The total score for each article was defined by the sum of the values obtained from four answers. A maximum value of 4.0 indicated a well-matched article for this scoping review, and a minimum of 0.0 indicated that the article was not suitable for this scoping review. A cutoff score of 2.0 was defined, so only articles scoring greater than 2.0 were accepted for this scoping review.

Table 6 shows the results of the quality assessment. The input set of this stage contained 26 articles, of which 8 scored less than or equal to 2.0. Thus, 18 articles passed to the next stage, the Data Extraction.

Results

A total of 705 articles were initially identified to examine IoT-based devices developed for individuals with visual impairment. After removing 267 duplicate records, 483 abstracts

Table 2: Database and search strings

Database	Search string	Number of articles	% of each search engine contributed
PubMed (PM)	("IoT AND Visually Impaired People") AND ("IoT OR assistive technology) OR ("IoT AND Visually impaired people")	73	10.35
IEEE Digital Library	("IoT and Assistive technology") AND (IoT, visually impairment) OR (IoT AND Visually impaired people")	40	5.67
Science Direct (SD)	("IoT and Assistive technology") AND (IoT, visually impairment) OR (IoT AND Visually impaired people")	398	56.45
ACM Digital	("IoT and Assistive technology") AND (IoT, visually impairment) OR (IoT AND Visually impaired people")	194	27.51
Total		705	100

Table 3: Inclusion criteria (IC)

ID	Criteria
IC1	Studies published between 2020 and 2025
IC2	Peer-reviewed original research articles
IC3	Studies within the context of IoT applied to AT
IC4	Articles published in English

Table 4: Exclusion criteria (EC)

ID	Criteria
EC1	Systematic review papers within the context of IoT applied to AT
EC2	Short articles, books, and gray literature
EC3	Duplicate study
EC4	Studies prior to 2020

Table 5: Questionnaire for quality evaluation[9]

ID	Question
QA1	Are the study objectives clearly defined?
QA2	Do the authors describe in detail the use of the IoT in the assistive product?
QA3	Did the study perform a well-described experiment to evaluate the proposal?
QA4	Do the findings of the study indicate a reliability relevant to it?

Table 6: Scoring of each selected articles based on Parsifal tool

ID	Subject	Author	Score
A1	Wearable smart glasses and an intelligent walking stick	[10]	3
A2	Application of smart objects	[11]	3.5
A3	Wearable glasses	[12]	4
A4	Stick	[13]	4
A5	Stick	[14]	3
A6	Smart glasses	[15]	4
A7	Stick	[16]	4
A8	Smart cane	[17]	4
A9	Wearable navigation device	[18]	2.5
A10	Smart cane	[19]	3
A11	Navigation system	[20]	2.5
A12	Real-time location	[21]	3
A13	OCR braille application	[22]	4
A14	Smart belt and smart cane	[23]	3
A15	YOLOv5s and Mask R-CNN	[24]	4
A16	Navigation	[25]	3
A17	Hybrid model	[26]	4
A18	Smart shoes	[27]	4

(OCR: optical character recognition; YOLO: you only live once; R-CNN: region-based convolutional neural networks)

were screened for relevance. Following screening, 26 articles were selected for the subsequent quality assessment stage and data extraction. Finally, 18 articles met the quality standards and reported the necessary information based on research questions that were included in the final review [Table 7].

The present scoping review highlights recent advancements in the IoT that have significantly influenced the development of ATs for individuals with visual impairments. The majority of IoT-integrated ATs are wearable smart and portable devices such as smart glasses, smart canes, belts, and shoes. These devices are embedded with sensors, microcontrollers, and cloud-based technologies to assist in navigation, object detection, and environmental interaction. For example, Chang *et al*.[10] developed smart glasses and an intelligent walking stick capable of identifying falls and obstacles using vibration feedback and a cloud-based Global Positioning System (GPS). Similarly, Connier *et al*.[11] proposed a smart object framework that facilitates interaction between blind users and their surrounding objects using Raspberry Pi and GPS. The results showed that several studies employed the Raspberry Pi and Arduino platforms, which offer cost-effective and flexible options for real-time processing. Masud *et al*., 2021[13] used a combination of Raspberry Pi 4B, camera, and ultrasonic sensors to enable object detection through a smart stick. Further, Pathak *et al*.[14] implemented a voice-controlled blind stick using Arduino Uno, presenting a low-cost yet basic solution with limited data processing capability. Cloud integration was also evident in studies like Achirei *et al*.[21] where GPS and cloud services were used to identify appropriate vehicles in public transportation systems.

Application of IoT in vision-assistive technology

This review showed that 38% of articles are related to IoT-enabled smart canes/sticks and 16% each to navigation devices and wearable smart glasses. A range of technologies has been developed to support individuals with visual impairment through the integration of IoT features. Various wearables, various sensors, smart canes, navigation devices, and Optical Character Recognition (OCR) Braille applications are vision-assistive devices in which IoT technology is incorporated [Table 8].

Further, voice-controlled virtual assistants, such as Google Assistant, Apple's Siri, and Amazon Alexa (Echo Dot), allow users to communicate with technology through natural language instead of visual or physical input. With just voice commands, users can utilize these systems to send messages, retrieve information, set reminders, operate smart home appliances, and even navigate their surroundings. Smart home ecosystems like Google Home and Amazon Echo further enhance independent living among visually challenged individuals by using audio cues to control lights, thermostats, appliances, and security systems.[28] People who are blind or visually impaired can now manage their daily activities with greater independence and self-assurance due to hands-free, speech-driven interaction. The increasing affordability and integration of these IoT-enabled tools into consumer technology highlight their potential as scalable, user-friendly assistive solutions for people with visual impairments.[29,30]

The findings also shared that despite the promising results, many studies reported significant limitations [Table 7]. Several devices were not tested on actual blind users[15] or lacked the ability to detect specific environmental features like stairs.[23,25]

Table 7: Research paper included in the present scoping review

References	IoT devices	Peripherals and Libraries Used	Results	Limitations/Drawbacks
[10]	Wearable Smart glasses and an Intelligent Walking stick	Vibration, Cloud based platform, GPS	Fall, Obstacle identification	Not integrated with deep learning
[11]	Application of Smart Objects	GPS, Raspberry Pi, uSu-EDU Board	Interface between user and its neighboring objects	Not able to exchange large quantity of data between VIP and objects
[12]	Wearable glasses	Raspberry Pi, Camera, Neural Network	Real-time object recognition	The accuracy of object detection and distance estimation is contingent upon the quality
[13]	Stick	Raspberry Pi-4B, Camera, Ultrasonic sensor, Arduino	Object detection	It identifies the object in particular situation regardless of frame
[14]	Stick	Arduino Uno	Voice control blind stick	--
[15]	Smart glasses	Arduino Uno, Buzzer feedback, GPS	Navigation	Testing was not done on blind and VIP usability of system is lacking
[16]	Stick	GPS, Raspberry Pi, Camera, speak vibration	Obstacle recognition	It is overcrowded, and navigation cannot be conducted
[17]	Smart Cane	Raspberry Pi	Object detection	Better real-time performance can be achieved by adopting newer raspberry Pi Model
[18]	Wearable Navigation Device	RGB-D, Camera	Navigation Device	--
[19]	Smart Cane	Micro controller, GPS, Camera, Raspberry Pi,	Object detection	---
[20]	Navigation system	WiFi, Deep hybrid Convolutional neural network	Indoor activity monitoring	---
[21]	Real time location	Cloud Service, GPS	Identify the correct vehicle in public transportation	--
[22]	OCR Braille application	Tesseract	Speech to text	Lack support for multiple languages
[23]	Smart belt and smart cane	Raspberry Pi 3, Earphone	Obstacle detection	Not able to detect stairs
[24]	YOLOv5s and Mask R-CNN	LiDAR, cloud service	Object detection	Heavy device
[25]	Navigation	Python's Text to speech	Navigation assistance	Not able to detect stairs
[26]	Hybrid Model	ESP32 Camera, Node MCU	Detect fresh fruits and vegetables	Limited dataset
[27]	Smart Shoes	GPS, GSM, Arduino Uno	Obstacle detection	---

Other systems faced issues like restricted datasets[26] (Oliullah *et al.*, 2025), limited language support[22] (Reddy *et al.*, 2024), or outdated hardware that compromised real-time performance.[17] Additionally, many IoT devices faced challenges such as bulkiness, high power consumption, or limited computational capabilities, which affected their usability and scalability. They often rely on stable Internet connectivity.

Functionality and accuracy of IoT-enabled AT

As shown in Table 9, the IoT devices utilized various technologies such as sensor-based technologies, machine and deep learning models, and hybrid AI (artificial intelligence) techniques.[31] Sensor-based approaches incorporate infrared, ultrasonic, Light Detection and Ranging (LiDAR), and motion tracking sensors to detect obstacles and assist with navigation. For example, smartphones and embedded platforms like Raspberry Pi or Arduino frequently serve as core processing units, linking various sensors such as ultrasonic transceivers, infrared cameras, or LiDAR modules.[32] Raspberry Pi boards are cost-effective and run on Linux-based operating systems like Raspberry Pi OS, allowing them to support multiple software tools, including programming languages such as Python, Java, and C++, as well as various IoT platforms and cloud-based services.[32]

Sensor-based technologies are foundational in many IoT-assisted navigation systems. Commonly used sensors include infrared sensors, which detect heat signatures; ultrasonic sensors, which measure the distance between objects using sound waves; and LiDAR, which monitors the user's movement and orientation.[24,33] These sensors enable real-time obstacle detection and path guidance, which are essential for safe mobility in dynamic environments. In addition to sensors, deep learning models such as convolutional neural networks (CNNs) and object detection

algorithms like you only look once (YOLO) and Mask R-CNN have been adopted for image recognition and scene understanding.[12,24] These models significantly enhance the accuracy of visual information processing by enabling devices to classify, locate, and interpret objects in the user's environment. Notably, Dhou *et al*.[19] achieved the highest accuracy (99–100%) for object identification using a combination of a six-axis accelerometer, gyroscope, and ultrasonic sensors. Deep learning-based methods, particularly CNNs, have also been used for visual data processing and obstacle recognition. Deepa *et al*.[20] reported 99% accuracy, while Khan *et al*.[24] used YOLO V5s and Mask R-CNN, achieving 97% accuracy.

Table 8: IoT-enabled visual-assistive devices

Primary Articles (ID)	Visual-assistive device
A1, A3, A6	Wearable smart glasses
A1, A4, A5, A7, A8, A10, A14	Smart stick/cane
A2	Application-based smart objects
A9, A11, A16	Navigation device
A12	Real-time location device
A13	OCR Braille application device
A14	Smart belt
A15	YOLO v5s
A17	Hybrid Model device
A18	Smart shoes

OCR - Optical character recognition, YOLO - You only look once

Furthermore, hybrid AI approaches, which combine traditional sensor inputs with machine learning outputs, are increasingly used to improve decision-making and adaptability of the devices.[20] These integrated systems contribute to a more intelligent and responsive assistive solution, helping users navigate complex real-world settings more effectively. These integrated technologies collectively enable diverse applications, including text recognition, facial identification, object tracking, and autonomous navigation, to fulfill the needs of users with vision loss.

Future research directions in IoT-enabled vision-assistive technology

As technology continues to grow, there are many opportunities to improve assistive devices for people with visual impairments. Based on the current research, there are a few important directions for future studies that could make IoT-enabled ATs more helpful, affordable, and user-friendly. First, more research is needed to make these devices easier to use in daily life. Many current tools can be too complex or bulky, which limits the usage among the end users. Future designs should focus on making devices smaller, more comfortable to wear, and easier to control.[31] Other than that, most of the existing devices are tested only in labs or in small pilot studies. There is a need for real-world testing with a larger number with a destination of users' choice.[34] In addition, more attention should be given to protecting users' privacy. Devices that use cameras, microphones, or GPS can collect sensitive information.[35] Furthermore, future studies should explore how to build low-cost devices.[36] Research on IoT-based AT for vision should aim for solutions that are user-friendly,

Table 9: Technological features and reliability of IoT-based vision-assistive devices

References	Technology used	Reliability (%)	Solution types
[10]	Infrared (IR) Transceiver Sensor Module, Motion tracking sensor module (MPU6050), Link It tm, LoRA LPWAN Module	98.3	Fall, Obstacle identification
[11]	SO2SEES system, IR/US Sensor	--	Interface between user and its neighboring objects
[12]	CNN	--	Real-time object recognition
[13]	Ultrasonic Sensor	91	Object detection
[14]	Infrared Sensor/ Ultrasonic Sensor	93	Voice control blind stick
[15]	LiDAR, Ultrasonic Sensor	96.49	Navigation
[16]	Ultrasonic Sensor, Water Sensor	--	Obstacle recognition
[17]	Ultrasonic Sensor	84.4	Object detection
[18]	SLAM and Semantic Segmentation Methods, CNN	--	Navigation Device
[19]	Six Axis accelerometer, Gyro ultrasonic Sensor	99-100	Object detection
[20]	CNN	99	Indoor activity monitoring
[21]	Maduino Zero 4G LTE	--	Identify the correct vehicle in public transportation
[22]	OCR	78(OCR) 85(Braille Translation Model)	Speech to text
[23]	Ultrasonic Sensor	--	Obstacle detection
[24]	YOLO V5s and MASK R CNN	97	Object detection
[25]	NVIDIA GeForce GTX1080 TiGPU	87.8	Navigation assistance
[26]	Hybrid deep learning, Efficient Net B7, VGG16, ResNet 50, VGG19	99.92	Detect fresh fruits and vegetables
[27]	Ultrasonic Sensor, Microcontroller	--	Obstacle detection

LoRA - long range, LPWAN - low power wide area network, OCR - optical character recognition, YOLO - you only look once, R-CNN - region-based convolutional neural networks

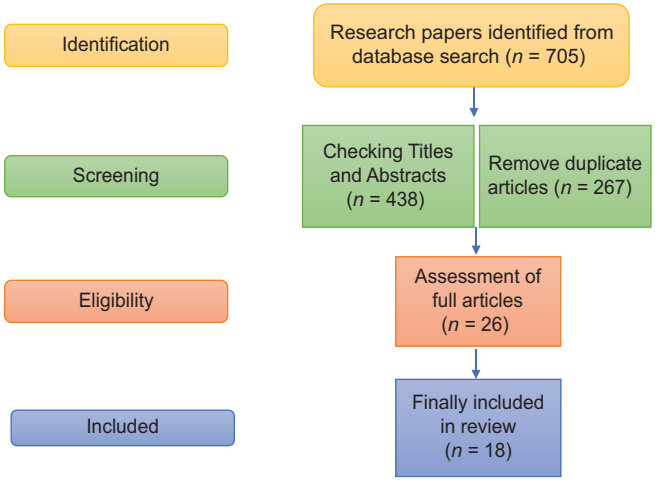


Figure 1: Flow chart depicting PRISMA flowchart for systematic screening of papers

well tested, secure, and affordable. The involvement of end-users and the focus on real-world challenges will be key to developing effective and inclusive tools.

Discussion

The rapid advancement in digital technologies, particularly the IoT, has brought significant transformation in the delivery of healthcare services. IoT-based systems are increasingly being leveraged to support disease prevention, monitoring, diagnosis, and treatment. These systems empower both healthcare providers and users to monitor health indicators such as heart rate, physical activity, and other vital signs remotely and in real time.[37] However, the utility of IoT extends much beyond general healthcare to specialized areas like its integration into assistive technologies for individuals with disabilities, including visual disabilities.

The present review highlights a range of IoT-enabled assistive devices are available that aim to improve mobility, independence, and environment interaction for individuals with visual impairments. Further, the accessibility of many devices has been greatly improved by the use of IoT technologies. These devices were often designed to be wearable, portable, and energy-efficient, ensuring that could be used throughout the day with minimal discomfort.

To be able to incorporate, the studies all across show a growing trend toward integrating technologies such as ultrasonic sensors, GPS, Raspberry Pi, Arduino microcontrollers, and AI-based libraries for object detection and navigation. Smart canes, wearable glasses, and smart belts are among the most commonly explored technologies. These devices use various peripherals to support real-time feedback, obstacle recognition, and user assistance through vibration alerts, audio cues, or haptic feedback. Studies such as those by Pathak *et al.*[14] and Dhou *et al.*[19] demonstrated successful implementation of voice control and real-time object detection, showcasing how embedded systems and speech interfaces can enhance user experience.

Assistive devices such as walking sticks, smart canes, smart glasses that provide auditory or haptic feedback, and mobile applications that utilize smartphone cameras and GPS for navigation assistance were tested by Ali and Khan *et al.*,[23,24] Various challenges were reported, such as the inability to detect stairs, limited support for multiple languages, and bulkiness of hardware components. These issues suggest a need for further innovation in form factor optimization and natural language processing to enhance usability. Additionally, several challenges need to be addressed in future research, such as hardware limitation, latency in data processing, high power consumption, and lack of integration with deep learning models for advanced environmental interpretation. Improving real-time rapid processing and accuracy remains crucial, especially for scenarios involving dynamic environments or crowded public spaces; therefore, future is warranted to minimize the challenges. Moreover, many current systems operate on limited data sets or in controlled environments, raising concerns about their performance in real-world settings.

Furthermore, future research should aim to enhance contextual awareness and adaptability in IoT-based ATs. Current devices primarily function through static commands or predefined scripts. Incorporating machine learning and AI can enable systems to better understand the user behavior, learn preferences over time, and deliver personalized responses. For example, a cane embedded with AI (IoT-enabled cane) could adapt its feedback mechanism based on terrain type, speed of movement, or ambient noise levels. Another area, affordability and accessibility, remains a significant concern. While the technologies exist, their adoption among underprivileged or rural populations is limited due to high cost and lack of availability. Strategies such as open-source development, modular hardware design, and community-driven innovation can help democratize access to these technologies. Supporting infrastructure such as reliable Internet connectivity and user training programs is also essential for widespread adoption.

It is also important to consider the ethical and privacy implications of IoT-based assistive devices. The continuous collection and transmission of personal data pose risks to privacy, security, and user autonomy. Issues related to data protection, service reliability, and device scalability have been reported across several studies. According to Shuai *et al.*,[38] ensuring secure data while handling and maintaining user trust are critical challenges in the implementation of IoT-based technology in healthcare and devices. In the end, the evolution of IoT-assisted technology for individuals with visual impairment marks a promising step toward inclusivity and independent living. While current devices offer valuable functionalities such as obstacle detection and navigation assistance, the field still faces technical, ethical, and economic challenges. Addressing these through interdisciplinary research, policy support, and user cantered design will be crucial to ensure that IoT-based ATs are not only innovative but also practical, scalable, and accessible to all.

Limitations of the study

Excluding articles which cannot be accessed or other non-English language may limit the representative to answer the research questions. Additionally, the selection of database or digital libraries presents threat since these may not cover the completeness of the studies carried out in the context of the topic.

Conclusion

Overall, the present scoping review demonstrates a growing trend toward intelligent, wearable, and IoT-integrated assistive devices for the visually impaired. It highlights the various

IoT-enabled solutions, including smart canes, wearable devices, navigation assistance, and environment sensing systems, that contribute to safer and more efficient navigation for users. Such devices play a transformative role in empowering visually challenged people. Nevertheless, challenges such as usability testing, hardware limitations, dataset constraints, and environmental adaptability remain. Addressing these issues in future research is essential to developing scalable and user-friendly assistive technologies. Future research should also focus on enhancing energy efficiency, reducing hardware costs, and improving AI-based decision making within IoT frameworks to make these devices more accessible.

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